

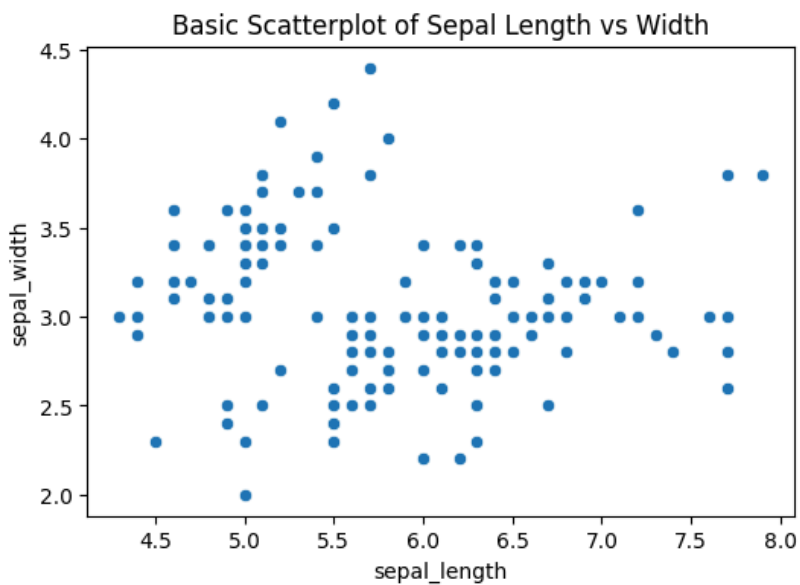
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
df = sns.load_dataset('iris')
```

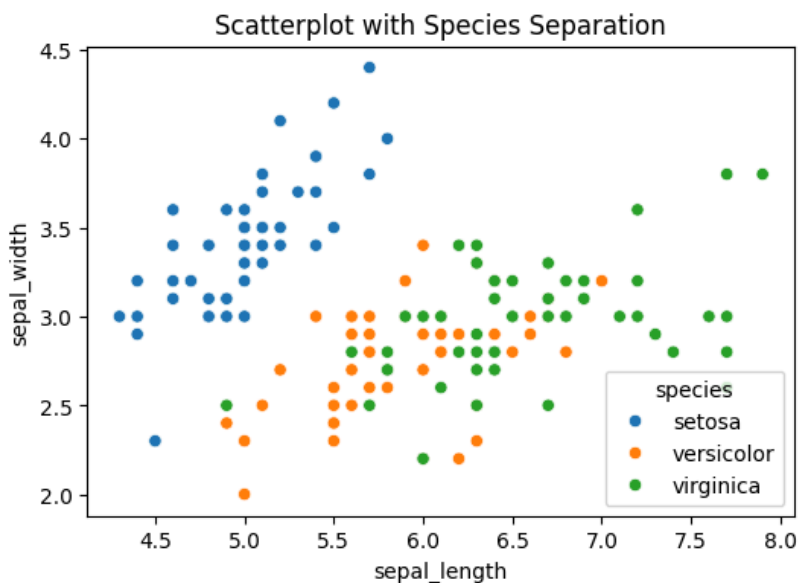
```
print(df.head())
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

```
plt.figure(figsize=(6, 4))
sns.scatterplot(x='sepal_length', y='sepal_width', data=df)
plt.title('Basic Scatterplot of Sepal Length vs Width')
plt.show()
```



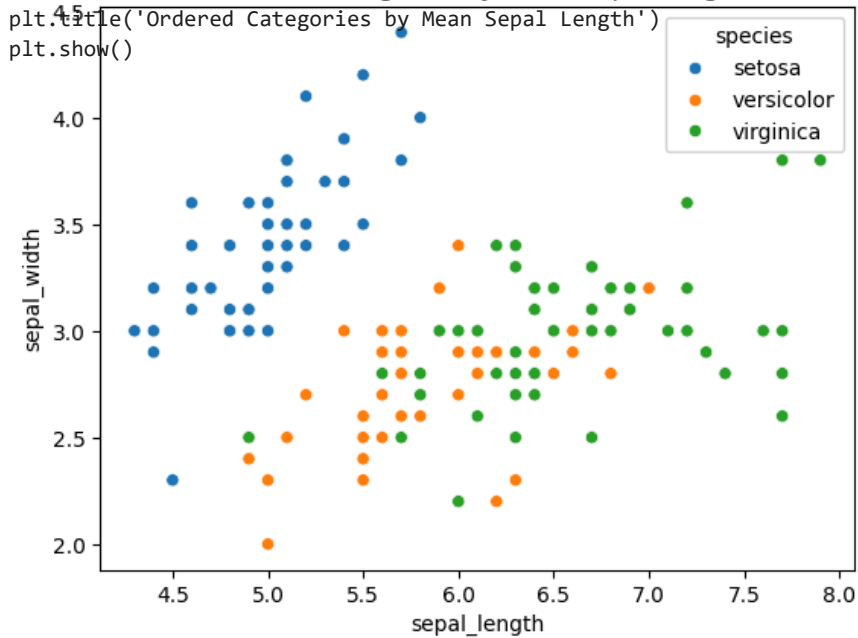
```
plt.figure(figsize=(6, 4))
sns.scatterplot(x='sepal_length', y='sepal_width', hue='species', data=df)
plt.title('Scatterplot with Species Separation')
plt.show()
```



```

order = df.groupby('species')['sepal_length'].mean().sort_values().index
sns.scatterplot(x='sepal_length', y='sepal_width', hue='species', data=df, hue_order=order)
plt.title('Ordered Categories by Mean Sepal Length')
plt.show()

```



```

g = sns.FacetGrid(df, col='species', height=4, aspect=1)
g.map_dataframe(sns.scatterplot, x='sepal_length', y='sepal_width')
g.set_titles(col_template="{col_name}")
g.fig.suptitle("Scatterplots Aligned by Species (Categorical Regions)", y=1.05)
plt.show()

```

